

Bucky++

Variant: 1.0

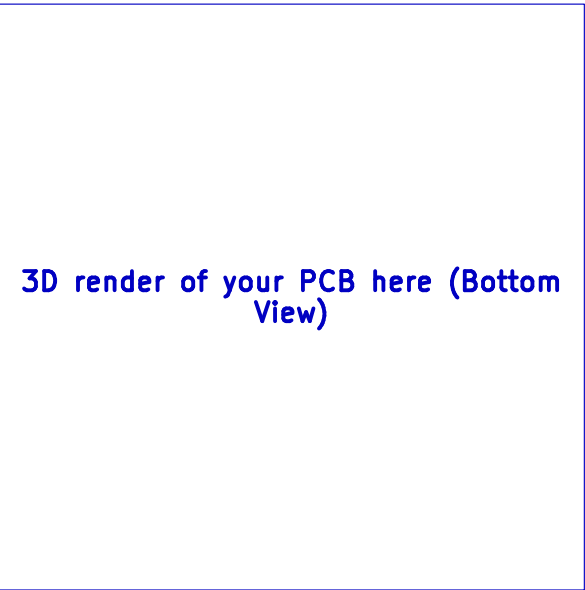
2025-11-19
Rev B

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TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

1.0 Release Date

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

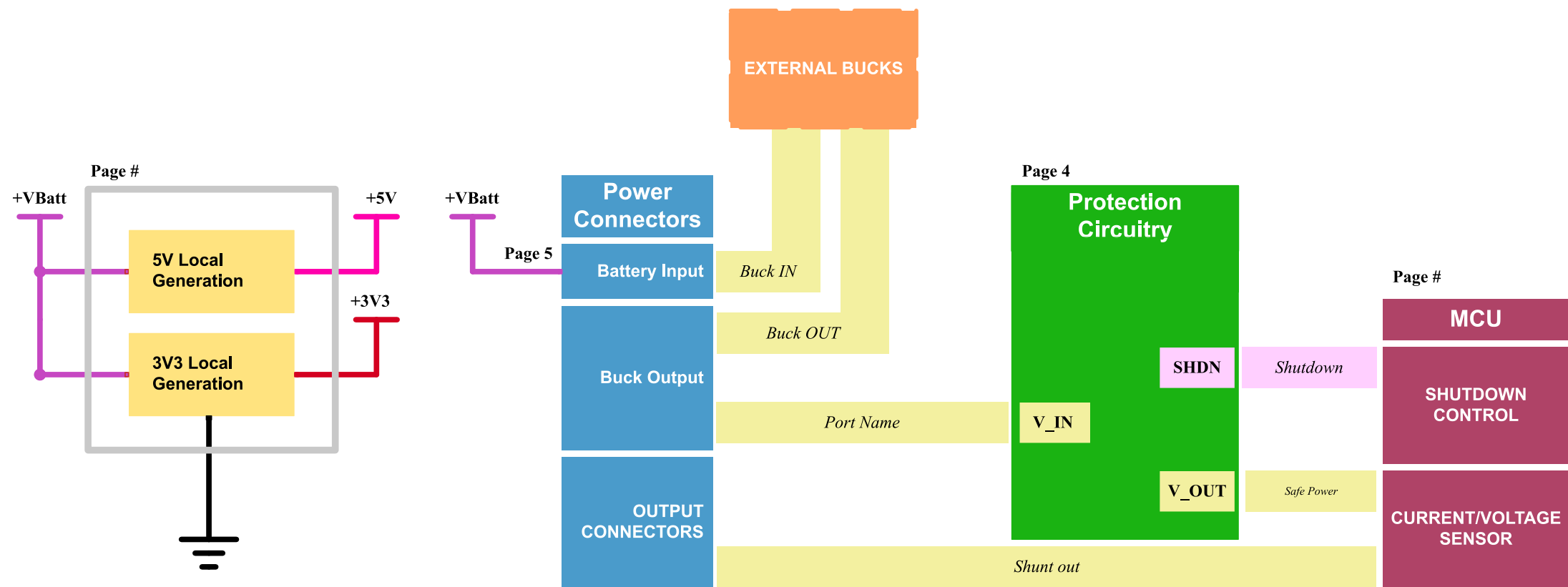
DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Cover Page	File Name: Bucky++.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
	Sheet Path: /		Reviewer: Reviewer Name	Size: A3	Sheet: 1 of 12

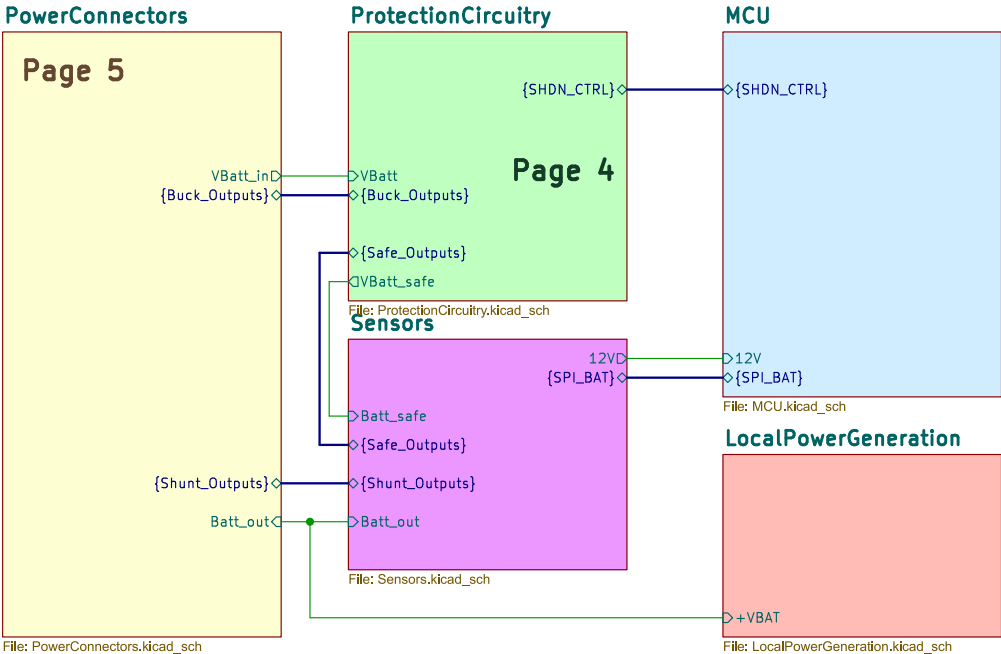
[2] Block Diagram



Target specifications:	
Input voltage:	22 - 26 V
Spec 2	??
Spec 3	??
Spec 4	??

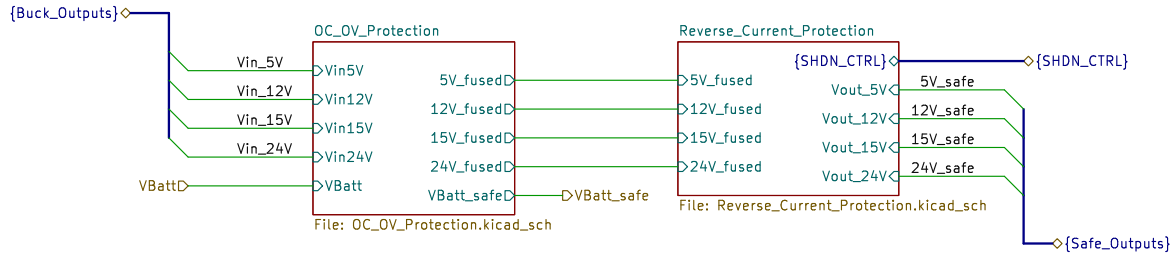
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		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Block Diagram/			Reviewer: Reviewer Name	Size: A3	Sheet: 2 of 12

[3] Project Architecture



	Comments:		Company: Xplore		Variant: 1.0	
			Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Project Architecture		File Name: Project Architecture.kicad_sch	Designer: Alain		Date: Last Modified Date
Sheet Path: /Project Architecture/			Reviewer: Reviewer Name		Size: A4	Sheet: 3 of 12

[4] Protection Circuitry



	Comments:		Company: Xplore		Variant: 1.0	
			Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Protection Circuitry		File Name: ProtectionCircuitry.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
	Sheet Path: /Project Architecture/ProtectionCircuitry/			Reviewer: Reviewer Name	Size: A5	Sheet: 4 of 12

[11] OC_OV_Protection

DESIGN NOTE:
ILIM is set at around
10A for all rails

DESIGN NOTE:
OV = 6V
UV = 4V

DESIGN NOTE:
OV = 13V
UV = 17V

DESIGN NOTE:
OV = 10V
UV = 14V

DESIGN NOTE:
OV = 22V
UV = 26V

Battery Hotswap controller

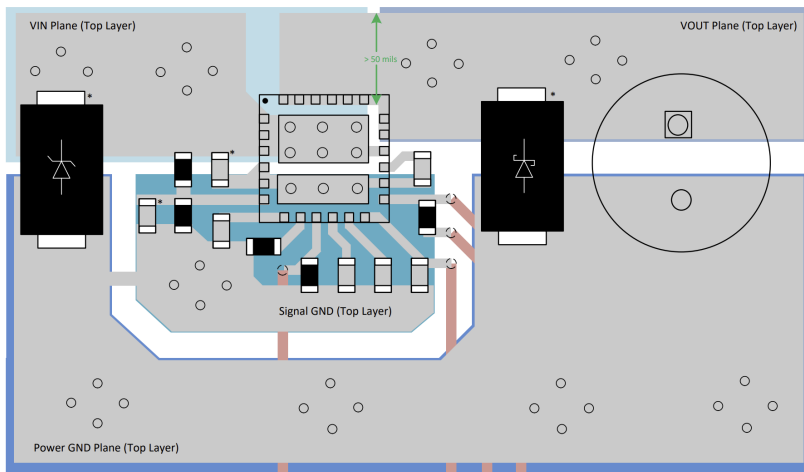
DESIGN NOTE:
A current slew rate can be set by limiting the input current at I_{INRUSH} = CL • 10µA/C26
With CL being the capacitance on the Nav Board (2*68µF = 136µF)
Here we set it at 0.62A.

A timer during which the max current is exceeded can be set using the formula:
C(nF) = 62 • t(ms)
Here we set it at 1ms

DESIGN NOTE:
A 1mOhm Shunt resistor is used to set the current limit at 47A (I=47mV/1mOhm).
Using a voltage divider, UV was set at roughly 22V (1.233V on the ON pin).

LAYOUT NOTE:

The minimum trace width for 1oz copper foil is 0.02" per amp to make sure the trace stays at a reasonable temperature. 0.03" per amp or wider is recommended.



* Optional components for suppressing transients induced while switching current through inductive elements at input/output

Table 7-5. NRETRY and RETRY_DLY Combination Examples

Auto Retry Delay	915 ms	416 ms	91.7 ms	9.3 ms	3 ms
RETRY_DLY Capacitor	22 nF	10 nF	2.2 nF	220 pF	68 pF
No. of Auto Retries	NRETRY Capacitor				
4	Open				
16	47 nF	22 nF	4.7 nF	1 nF	220 pF
64	0.22 µF	0.1 µF	22 nF	2.2 nF	1 nF
256	1 µF	0.47 µF	0.1 µF	10 nF	4.7 nF
1024	3.3 µF	1.5 µF	0.47 µF	33 nF	10 nF
Infinite	Short to GND				

DESIGN NOTE:
The number of time the e-fuse can retry is set at 16 and the delay between retries is maximised by using the capacitor values in the table provided by TI.
IMON pin is not used as we already have sensing, it is wired as recommended in the datasheet.
The duration of overcurrent can be set using the capacitor connected to ITimer. We want the shortest possible duration so we leave the pin open

$$VIN(UV) = \frac{VUVLO(R) \times (R1 + R2 + R3)}{R2 + R3}$$

$$VIN(OV) = \frac{VOV(R) \times (R1 + R2 + R3)}{R3}$$

$$R_{ILIM}(\Omega) = \frac{1460}{ILIM(A) - 0.11}$$

$$t_{ITIMER}(ms) = \frac{C_{ITIMER}(nF) \times \Delta V_{ITIMER}(V)}{I_{ITIMER}(\mu A)}$$

Comments:

Company:

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1.0

Board Name:

Bucky++

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XRC

Sheet Title:

OC_OV_Protection

File Name:

OC_OV_Protection.kicad_sch

Designer:

Alain

Date:

Revision:

Sheet Path:

/Project Architecture/ProtectionCircuitry/OC_OV_Protection/

Reviewer:

Reviewer Name

Size:

A3

Sheet:

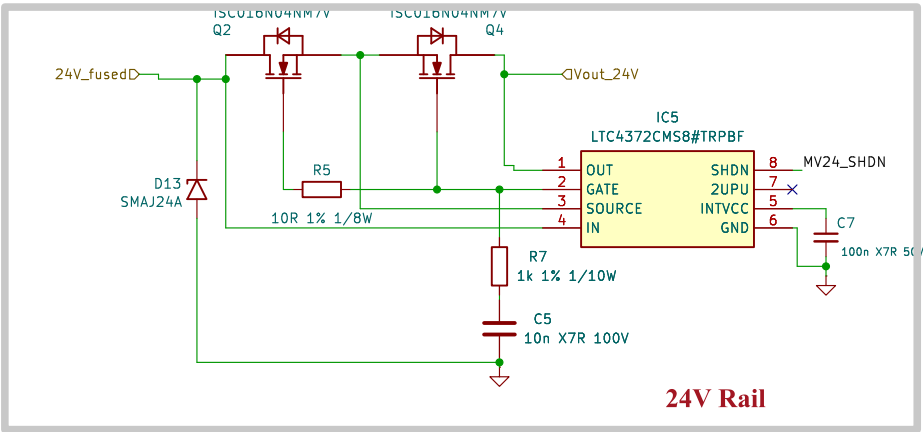
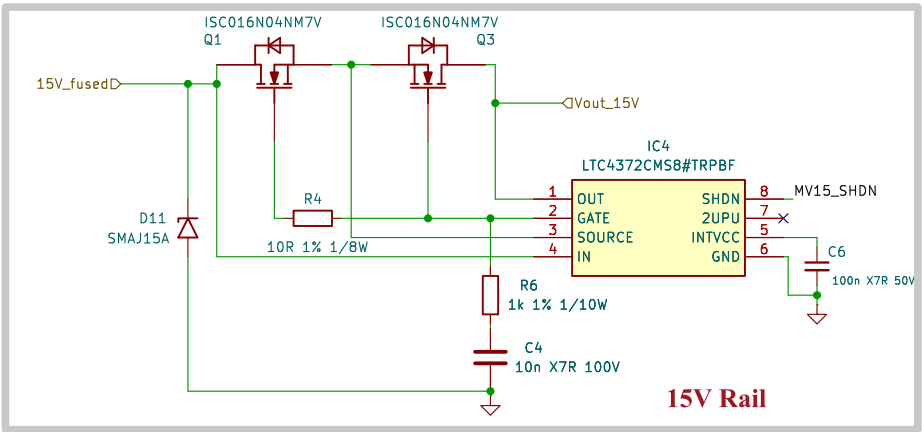
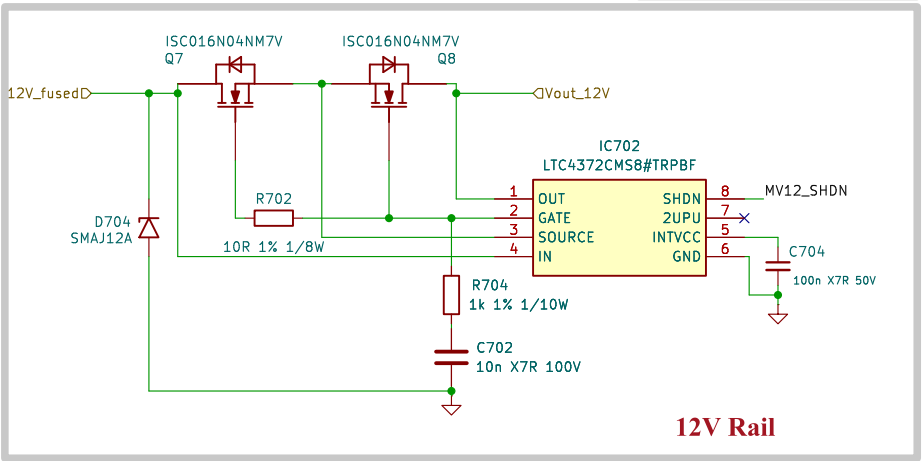
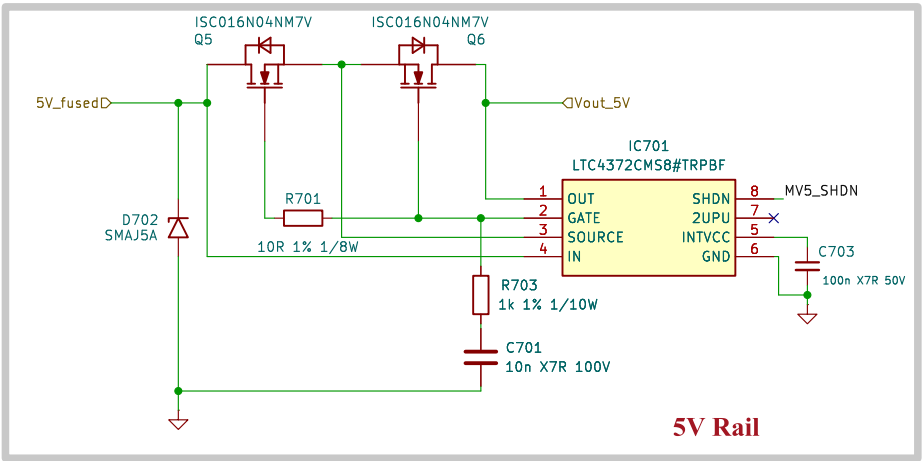
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[12] Reverse_Current_Protection

DESIGN NOTE:
TVS diodes are added to protect the chip
and MOSFETS

DESIGN NOTE:
ISC016N04NM7V was chosen as it can
hold 40V and up to 174A.

With power dissipation of over 80W, this is
more than enough even for the main input
as $P=I^2R$. This Transistor can theoretically
hold over 200A.



MV5_SHDN
MV12_SHDN
MV15_SHDN
MV24_SHDN
{SHDN_CTRL}

Comments:

Company:

Variant:

1.0

Board Name:

Bucky++

Project Name:

XRC

Sheet Title:

Reverse_Current_Protection

File Name:

Reverse_Current_Protection.kicad_sch

Designer:

Alain

Date:

Revision:

Sheet Path:

/Project Architecture/ProtectionCircuitry/Reverse_Current_Protection/

Reviewer:

Reviewer Name

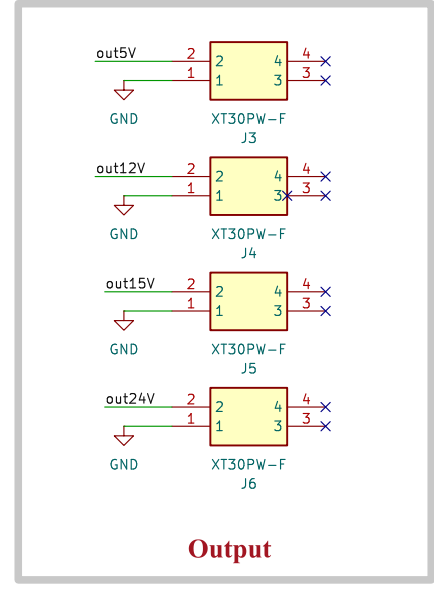
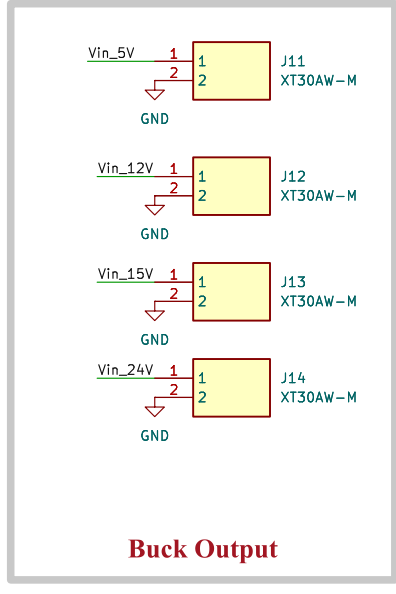
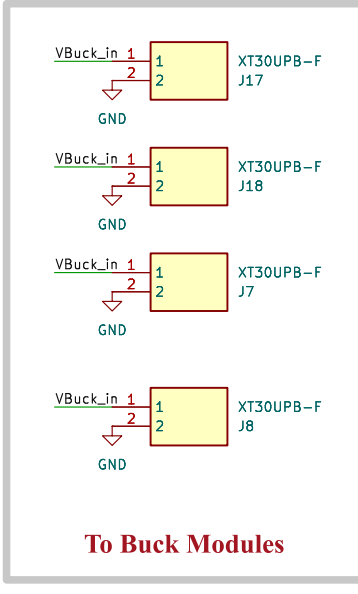
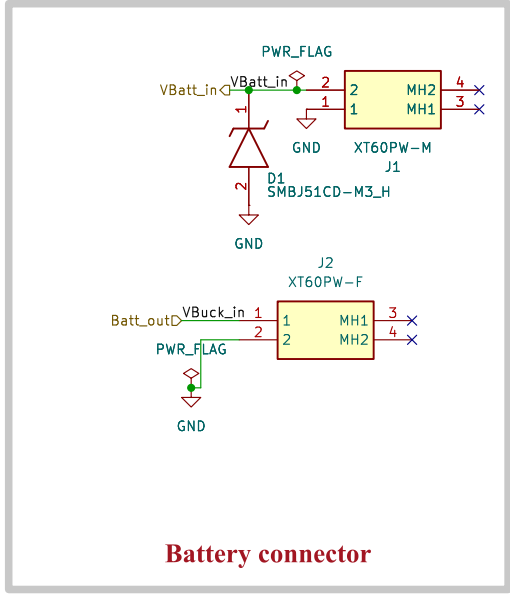
Size:

A4

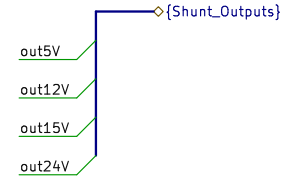
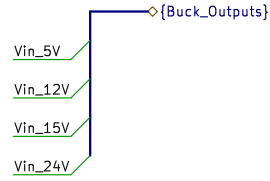
Sheet:

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[5] Power Connectors

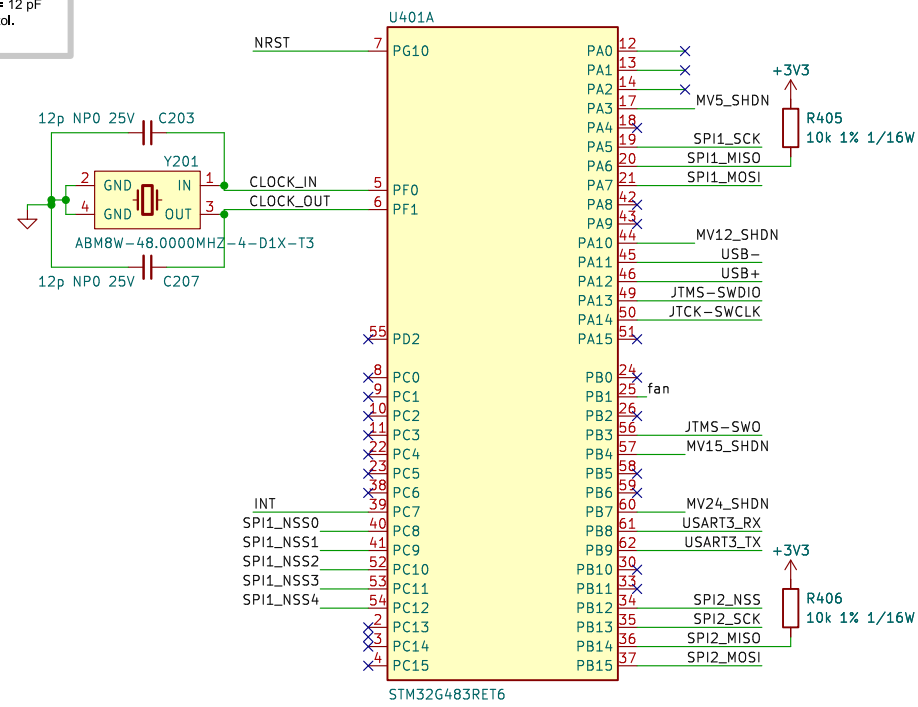


DESIGN NOTE:
The battery is directly connected to Bucky via J1 and the power is directly transferred to the motors through J2. It is also connected to the bucks' input through VBuck_in.
The output of the buck are then connected to their respective voltage nets.
The same ground is used for the whole board.



	Comments:	Company: Xplore		Variant: 1.0	
		Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Power Connectors	File Name: PowerConnectors.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Project Architecture/PowerConnectors/			Reviewer: Reviewer Name	Size: A4	Sheet: 5 of 12

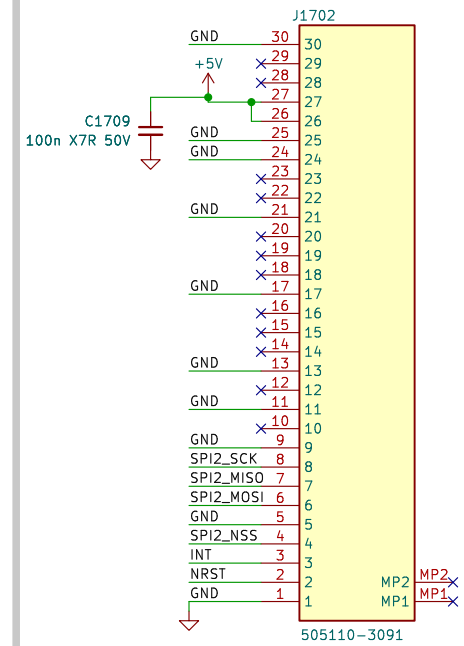
DESIGN NOTE:
Cs = 2 pF
CL = 8 pF
CL1 = CL2 = 2*(CL-CS) = 12 pF
Choose C0G/NP0 1-2% tol.



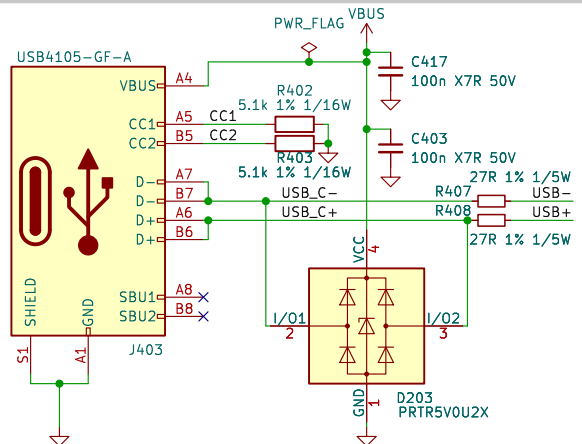
MCU I/O Domain

[8] MCU

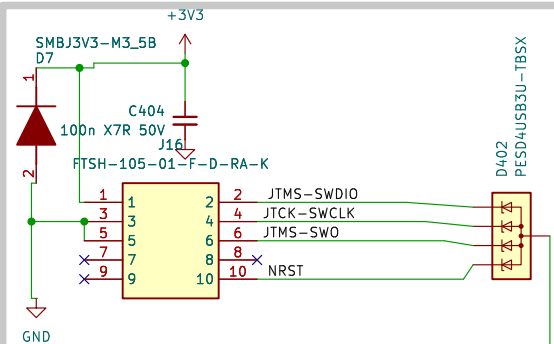
STM32 MCU that controls the power and communicate sensor data. The same model and same pins are used as GEODE to facilitate coding and developpement. Only the most important features are used here (Sensors, display, fan control, debug and basic communication connectors)



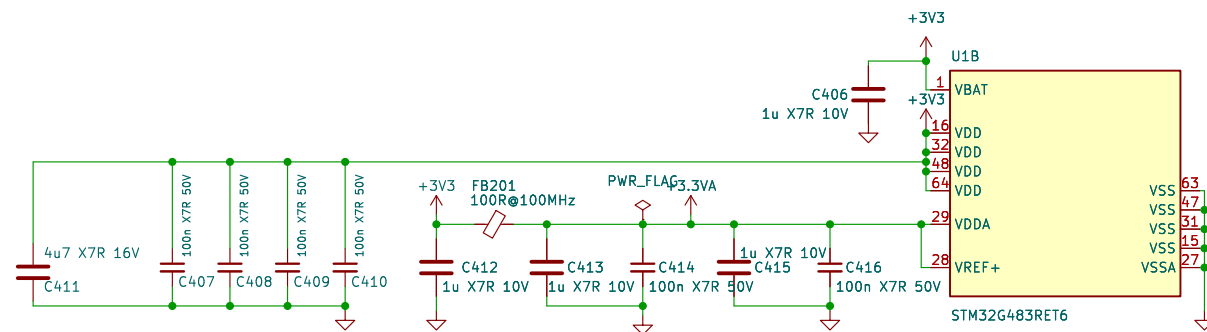
External Connector



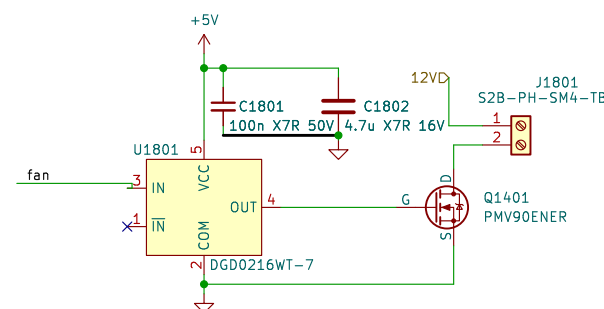
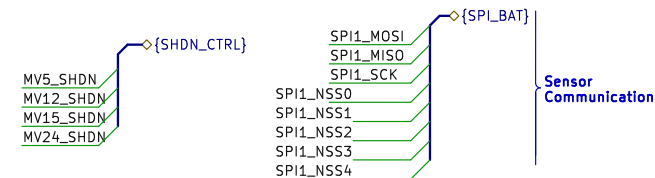
USB connector



SWD Debug Connector

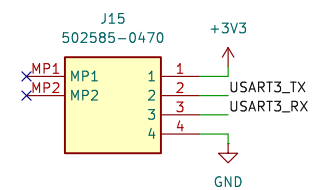


MCU Power Domain



FAN

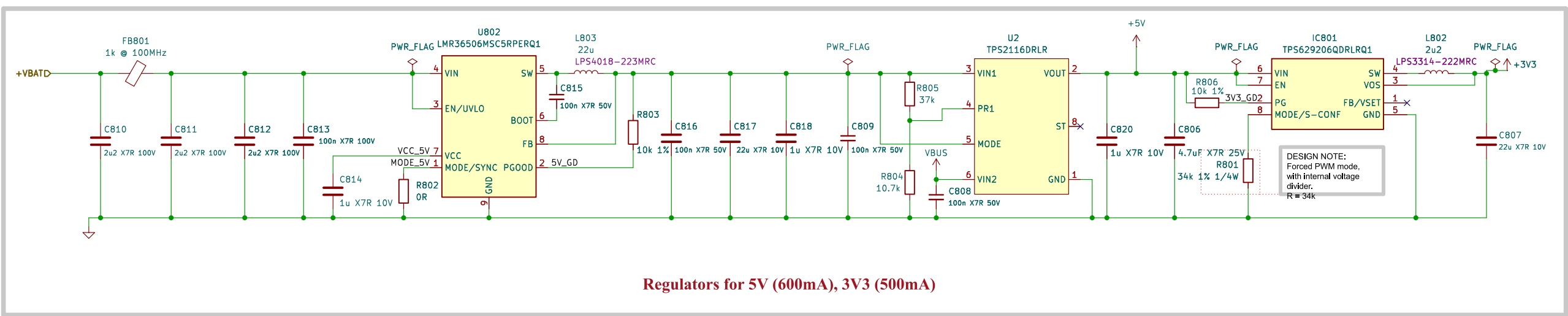
DESIGN NOTE:
The Fan is
powered by the
12V output from
the Bucks



USART Connector

Comments:	Company: Xplore		Variant: 1.0	
	Board Name: Bucky++		Project Name: XRC	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
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[9] Local Power Generation

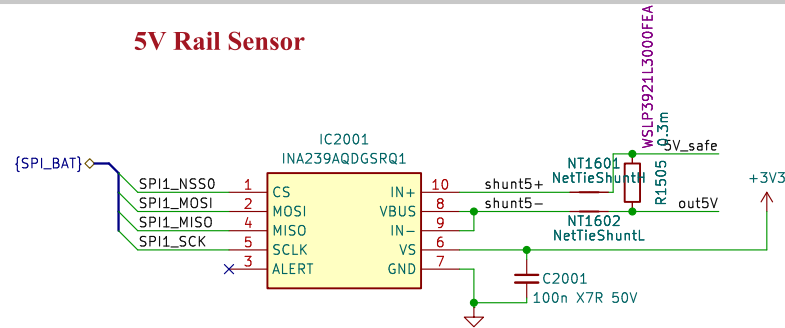


Local Power generation to power the MCU and peripherals. Schematics adapted from GEODE

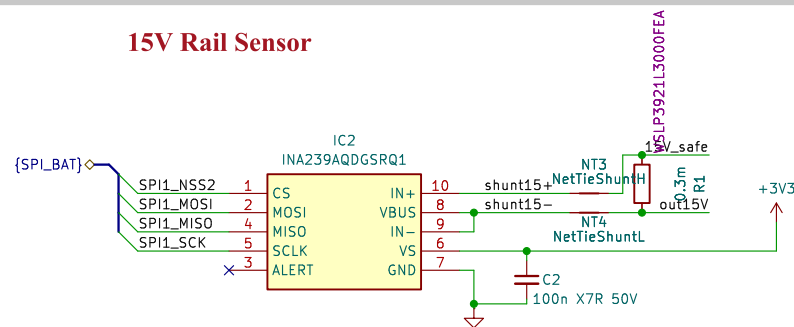
	Comments: References: Flexible I/O worked examples Flexible I/O source configuration		Company: Xplore		Variant: 1.0	
			Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Local Power Generation		File Name: LocalPowerGeneration.kicad_sch		Designer: Alain	
					Date: Last Modified Date	
				Revision: B		
Sheet Path: /Project Architecture/LocalPowerGeneration/			Reviewer: Reviewer Name		Size: A3	Sheet: 9 of 12

[10] Sensors

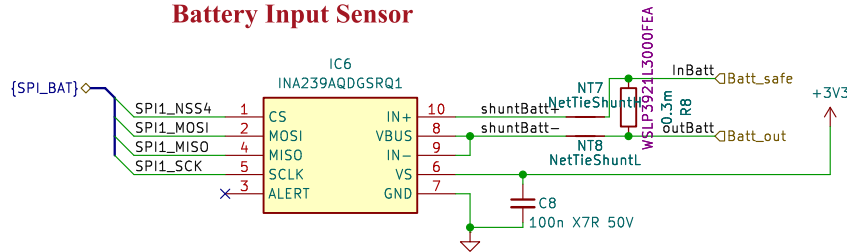
5V Rail Sensor



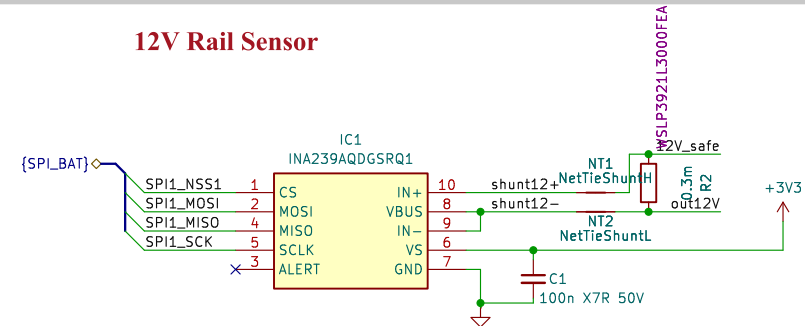
15V Rail Sensor



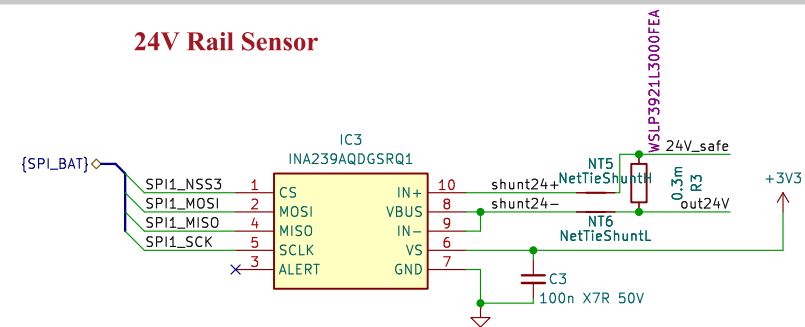
Battery Input Sensor



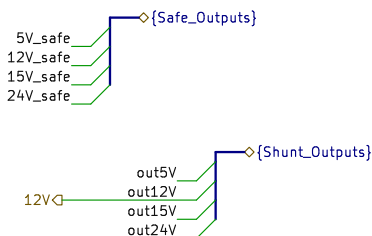
12V Rail Sensor



24V Rail Sensor



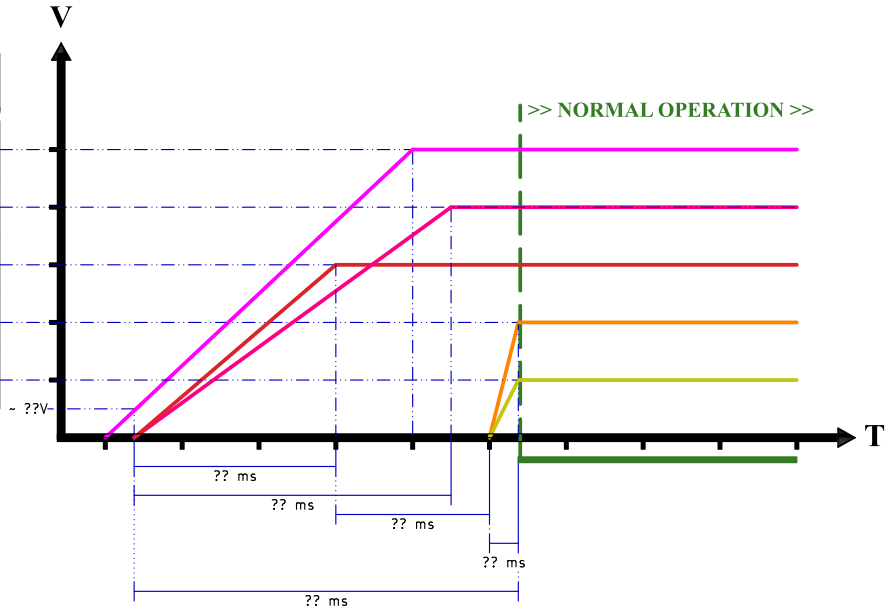
DESIGN NOTE:
One sensor for each rail is added to monitor current and voltage of the bucks' output.
One sensor is added for the input, it will help us deduce the motor power consumption. They all run on the same SPI as 8 uS is a low enough update interval. The update interval is computed by this formula:
Number of devices * bit / sensor frequency = 8 uS



Comments:	Company: Xplore		Variant: 1.0	
	Board Name: Bucky++		Project Name: XRC	
Sheet Title: Sensors	File Name: Sensors.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
Sheet Path: /Project Architecture/Sensors/		Reviewer: Reviewer Name	Size: A4	Sheet: 10 of 12

[6] Power - Sequencing

NAME	SOURCE	LEVEL
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%



	Comments:		Company: Xplore		Variant: 1.0	
			Board Name: Bucky++		Project Name: XRC	
	Sheet Title: Power - Sequencing		File Name: Power - Sequencing.kicad_sch	Designer: Alain	Date: Last Modified Date	Revision: B
	Sheet Path: /Power - Sequencing/			Reviewer: Reviewer Name	Size: A4	Sheet: 6 of 12

[7] Revision History

15.11.2025 - 1.0 Revision
Variant: A

- Added schematics and e-fuse for rail outputs
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	Comments:		Company:		Variant:	
			Xplore		1.0	
			Board Name:		Project Name:	
			Bucky++		XRC	
	Sheet Title:		File Name:		Designer:	
	Revision History		Revision History.kicad_sch		Alain	
	Sheet Path:		Reviewer:		Size:	
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